Math 55 Final Exam, Prof. Srivastava
December 15, 2022, 11:40am–2:30pm, 2050 VLSB.

Name: ______________________________

SID: ______________________________

GSI: ______________________________

Name of the student to your left: ______________________________

Name of the student to your right: ______________________________

Instructions: Write all answers clearly in the provided space. **Do not unstaple the exam.** Write your name and SID on every page. Write proofs in complete English sentences. To cite a theorem proved in class, just state it. **Be sure to fill in the multiple choice circles clearly.** Ambiguous multiple choice answers will be given a zero. Graph theory definitions are provided on the last page. You may use one 2−sided cheat sheet.

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Sign here: ______________________________

Question	Points
1	12
2	12
3	12
4	12
5	12
6	12
7	14
8	14
Total:	100

Do not turn over this page until your instructor tells you to do so.

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1. Select true or false for each of the following. There is no need to provide an explanation.

(a) (2 points) The propositions $\exists x(P(x) \lor Q(x))$ and $\exists x P(x) \lor \exists x Q(x)$ are logically equivalent.

● True ○ False

> **Solution:** True. Proof: $(\rightarrow)$. Assume the first proposition. By existential instantiation, we can choose x such that $P(x) \lor Q(x)$. If $P(x)$ then by existential generalization, $\exists x P(x)$. Otherwise, $Q(x)$ so $\exists x Q(x)$. Thus, the second proposition is true.
> $(\leftarrow)$. Assume the second proposition. Thus, either $\exists x P(x)$ or $\exists x Q(x)$. In the first case, choose x such that $P(x)$. Consequently, $P(x) \lor Q(x)$. By existential generalization, $\exists x P(x) \lor Q(x)$. The second case $\exists x Q(x)$ is analogous.

(b) (2 points) If n and k are positive integers with $k < n$, the function

$$f : \mathcal{P}(\{1, \dots, n\}) \to \mathcal{P}(\{1, \dots, k\})$$

defined by $f(S) = S \cap \{1, \dots, k\}$ is onto.

● True ○ False

> **Solution:** True. Assume $A \in \mathcal{P}(\{1, \dots, k\})$ is arbitrary. Since $\{1, \dots k\} \subseteq \{1, \dots, n\}$, we also have $A \in \mathcal{P}\{1, \dots, n\}$. By the definition of f, $f(A) = A$, as desired.

(c) (2 points) If n and k are positive integers with $k < n$, the function

$$f : \mathcal{P}(\{1, \dots, n\}) \to \mathcal{P}(\{1, \dots, k\})$$

defined by $f(S) = S \cap \{1, \dots, k\}$ is one to one.

○ True ● False

> **Solution:** False. Assume $k < n$. Observe that $f(\{1, \dots, n\}) = \{1, \dots k\}$ and $f(\{1, \dots, (n-1)\} = \{1, \dots, k\}$ since $k \leq n - 1$. Thus f is not one to one.

(d) (2 points) The set

$$S = \{T : T \text{ is a tree with vertex set } \{1, \dots, n\} \text{for some integer } n \geq 1\}$$

is uncountable.

○ True ● False

Solution: False. Let S_n be the set of all trees with vertex set $\{1, \ldots, n\}$; note that S_n is finite since the set of simple graphs on a given set is finite. Consider the sequence:

$$\text{elements of } S_1, \text{elements of } S_2, \text{elements of } S_3, \ldots$$

This is an enumeration of the set S since for every $n \geq 1$ every tree with vertex set $\{1, \ldots, n\}$j appears exactly once (during the enumeration of elements of S_n). Thus S is countable.

(e) (2 points) If n and r are positive integers with $r < n$ then

$$\binom{n+1}{r+1} = \binom{n}{r+1} + \binom{n}{r}.$$

● True ○ False

Solution: True. This is just Pascal's identity with the substitution $n+1$ instead of n and $r+1$ instead of r.

(f) (2 points) If X and Y are independent Geometric random variables both with parameter $q = 1/3$, then $X + Y$ is a Geometric random variable with parameter $q = 2/3$.

○ True ● False

Solution: False. There are many ways to see this, but one simple one is that $\mathbb{E}X = \mathbb{E}Y = 3$ so by linearity of expectation $\mathbb{E}(X + Y) = 6$. If the statement was true however, we would have $\mathbb{E}(X + Y) = 1/(2/3) = 3/2$, which is a contradiction.

2. Suppose that Nikhil is hungry, and selects a fruit by first picking one of two boxes labeled A and B uniformly at random, and then selecting a fruit uniformly randomly from that box. Box A contains 3 peaches, 3 apples, and 2 mangoes, and Box B contains 1 peach and 3 apples (and no mangoes).

   (a) (6 points) Given that Nikhil selected a peach, what is the probability that he chose Box A?

   ○ 1/2 ○ 3/8 ● 3/5 ○ 1/3
   ○ none of the above.

   Solution: Let P be the event that Nikhil selects a peach and let A be the event that he selects Box A (so that $\overline{A}$ is the event that he selects Box B). The question is asking for $\mathbb{P}(A|P)$. The description in the question tells us that:

   $$\mathbb{P}(A) = \mathbb{P}(\overline{A}) = 1/2 \qquad \mathbb{P}(P|A) = 3/8 \qquad \mathbb{P}(P|\overline{A}) = 1/4.$$

   By the law of total probability, this implies

   $$\mathbb{P}(P) = (1/2)\mathbb{P}(P|A) + (1/2)\mathbb{P}(P|\overline{A}) = \frac{3}{16} + \frac{1}{8} = \frac{5}{16}.$$

   By Bayes' rule, we have

   $$\mathbb{P}(A|P) = \mathbb{P}(P|A)\mathbb{P}(A)/\mathbb{P}(P) = (3/8)(1/2)/(5/16) = 3/5.$$

   (b) (3 points) If Nikhil selected a mango, what is the probability that he chose Box B?

   ○ 1/2 ○ 3/12 ● 0 ○ 1
   ○ none of the above.

   Solution: Let M be the event that a mango is selected. We have $\mathbb{P}(M|\overline{A}) = 0$ since there are no mangos in Box B. Thus,

   $$\mathbb{P}(\overline{A}|M) = \mathbb{P}(M|\overline{A})\mathbb{P}(\overline{A})/\mathbb{P}(M) = 0 \times \mathbb{P}(\overline{A})/\mathbb{P}(M) = 0.$$

   (c) (3 points) Are the events "Nikhil selected Box A" and "Nikhil selected a Peach" independent?

Solution: We have $\mathbb{P}(A) = 1/2$ and $\mathbb{P}(A|P) = 3/5$ as well as $\mathbb{P}(P) > 0$ by part (a). Since these probabilities are not equal, the events A and P are *not independent*.

○ Yes ● No

3. Suppose I throw 10 balls independently and uniformly at random into 5 bins labeled A, B, C, D, E. Call a bin *special* if it contains exactly 2 balls.

(a) (4 points) The expected number of special bins is:

○ $\frac{10!}{32\times 5^{10}}$ ● $\frac{5\times 45\times 4^8}{5^{10}}$ ○ $\frac{5\times 2\times 9^8}{10^{10}}$ ○ $\frac{5\times 10\times 4^8}{5^8}$

○ none of the above.

Solution: Let X be a random variable equal to the number of special bins. Let X_i for $i = 1, \ldots 5$ be an indicator random variable which is 1 iff the ith Bin (in A, B, C, D, E) is special. By linearity of expectation

$$\mathbb{E}X = \sum_{i=1}^{5} \mathbb{E}X_i.$$

Each term in the sum is the probability that a single bin is special. We calculate

$$\mathbb{E}X_1 = \mathbb{P}(\text{Bin A is special}) = \mathbb{P}(\text{Bin A contains exactly 2 balls}).$$

The number of balls in Bin A has a Binomial$(10, 1/5)$ distribution since there are 10 independent trials (each corresponding to throwing a ball) and the probability of success (landing in Bin A) is $1/5$ since there are 5 bins. This probability is $\binom{10}{2}(4/5)^8(1/5)^2 = 45\frac{4^8}{5^{10}}$, so $\mathbb{E}X_1 = 45\frac{4^8}{5^{10}}$. As the same calculation applies to each of the 5 bins, we have

$$\mathbb{E}X = 5 \times 45 \times \frac{4^8}{5^{10}}.$$

(b) (3 points) The probability that every bin is special is:

○ $(2/5)^5$ ○ $(\frac{45\times 4^8}{5^{10}})^5$ ● $\frac{10!}{32\times 5^{10}}$ ○ $1 - \frac{5\times 120\times 4^7}{5^{10}}$

○ none of the above.

Solution: Every bin is special if and only if every bin contains exactly two balls. The number of ways of placing 10 labeled balls into 5 bins is $\frac{10!}{2!2!2!2!2!}$ (since the number of such arrangements corresponds to a permutation of 10 balls, and then we apply the division rule to forget the order inside each bin). The total number of ways of placing 10 balls in 5 bins is 5^{10}. Thus, the probability that every bin is special is

$$\frac{10!}{32 \times 5^{10}}.$$

(c) (3 points) The probability that there exists a bin containing *at least two balls* is:

○ $1 - \frac{5 \times 45 \times 4^8}{5^{10}}$ ○ $2/5$ ○ $\frac{5 \times 2^5}{5^5}$ ● 1
○ none of the above.

Solution: By the pigeonhole principle, there must be at least one bin with $\lceil 10/5 \rceil = 2$ balls. Thus the probability that there exists a bin containing at least 2 balls is 1.

(d) (2 points) The events "Bin A is special" and "Bin B is special" are:

○ independent ● not independent

Solution: As calculated in part (a), the probability that Bin B is special is $45 \times 4^8/5^{10}$. The conditional probability $\mathbb{P}(\text{Bin } B \text{ is special}|\text{Bin } A \text{ is special})$ is calculated as follows: given that Bin A is special, the experiment consists of placing 2 balls in Bin A and throwing the remaining 8 balls independently and uniformly at random into the remaining 4 bins. The number of balls in Bin B in this new experiment is a Binomial$(1/4, 8)$, so

$$\mathbb{P}(\text{ Bin } B \text{ is special}|\text{Bin } A \text{ is special}) = \binom{8}{2}(3/4)^{6}(1/4)^2 = 28 \times 3^{6}/4^{8}.$$

since this is not equal to $\mathbb{P}(\text{Bin } B \text{ is special})$, the events are *not* independent.
This problem can also be done by direcly computing

$$\mathbb{P}(\textit{Bin } A \textit{ is special and Bin } B \textit{ is special})$$

using a counting argument.

4. The Math 555 final has 99 multiple choice questions numbered #1 − #99, each with 3 choices, *exactly one* of which is correct. The grading scheme is: every correct answer receives +1 point and every wrong answer receives −1/2 point. I have not studied at

all, so I answer all of the questions independently and uniformly at random. Let X be the random variable equal to the total number of points I earn on the exam.

(a) (3 points) The expectation of X is:

○ 99/3 ○ 1/3 ● 0 ○ 99/6
○ none of the above.

Solution: Let X_i be equal to my score on question i, for $i = 1, \dots, 99$. For each i, we have

$$\mathbb{E}X_i = (1/3)(+1) + (2/3)(-1/2) = 1/3 - 1/3 = 0.$$

By linearity of expectation,

$$\mathbb{E}X = \sum_{i=1}^{99} \mathbb{E}X_i = \sum_{i=1}^{99} 0 = 0.$$

(b) (3 points) The variance of X is:

○ $(99/3)^2$ ● 99/2 ○ 99(5/9) ○ 99/6.
○ none of the above.

Solution: For each $i = 1, \dots, 99$, we have

$$V(X_i) = \mathbb{E}X_i^2 - (\mathbb{E}X_i)^2 = \mathbb{E}X_i^2 - 0$$

by part (a). We calculate:

$$\mathbb{E}X_i^2 = (1/3)(+1)^2 + (2/3)(-1/2)^2 = 1/3 + 1/6 = 1/2.$$

Since the X_i are independent and $X = \sum_{i=1}^{99} X_i$, we have

$$V(X) = \sum_{i=1}^{99} V(X_i) = 99/2.$$

(c) (4 points) A *streak* is a sequence of 3 questions with adjacent numbers (e.g., 1,2,3 or 13,14,15) all of which I answered correctly. The expected number of streaks is:

○ $\frac{99 \times 98 \times 97}{6} \times 1/27$ ● 97/27 ○ 97/3 ○ 33

◯ none of the above.

Solution: Notice that there are 97 possible streaks in the exam:

$$(1,2,3),(2,3,4),\ldots,(97,98,99).$$

Let Y be a random variable equal to the total number of streaks. Let Y_i for $i=1,\ldots,97$ be the indicator random variable equal to 1 iff $(i,i+1,i+2)$ is a streak. Since the answers to the questions are independent, we have

$$\mathbb{E}Y_i = \mathbb{P}(i,i+1,i+2 \text{ are correct}) = (1/3)^3 = 1/27.$$

By linearity of expectation,

$$\mathbb{E}Y = \sum_{i=1}^{97} \mathbb{E}Y_i = 97 \times 1/27 = 97/27.$$

(d) (2 points) The events "I answer question #1 correctly" and "I answer question #99 correctly" are:

● independent ◯ not independent

Solution: Since the question states that I answer the questions independently, these two events are independent.

5. (a) (4 points) A value of x for which $55x \equiv 5(\textbf{ mod } 12)$ is:

◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7
◯ 8 ◯ 9 ◯ 10 ● 11 ◯ none of the above.

Solution: It is sufficient to find an x such that $11x \equiv 1(\textbf{ mod } 12)$ since then we can just multiply both sides of that congruence by 5 and x satisfies the given congruence. Since 11 and 12 are relatively prime, 11 must have an inverse modulo 12; to find it we use the Euclidean algorithm:

$$12 = 1 \times 11 + 1 \qquad 1 = 12 + (-1)11,$$

so $-1 \equiv 11(\textbf{ mod } 12)$ is an inverse of $11(\textbf{ mod } 12)$. Multiplying both sides of the congruence by 11, we have

$$11 \cdot 11x \equiv 11(\textbf{ mod } 12) \Rightarrow x \equiv 11(\textbf{ mod } 12).$$

Thus $x = 11$ is a valid answer (and in fact the only correct one, since inverses are unique).

An alternate approach is to reduce the equation modulo 12 to obtain the equivalent congruence $7x \equiv 5(\mod 12)$ and proceed from there.

(b) (4 points)] A value of y for which $54y \equiv 5(\mod 12)$ is:

○ 0 ○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6 ○ 7
○ 8 ○ 9 ○ 10 ○ 11 ● none of the above.

Solution: Since $54 \equiv 6(\mod 12)$ the congruence is equivalent to

$$6y \equiv 5(\mod 12).$$

This looks fishy since $gcd(6, 12) = 6$ so 6 does not have an inverse modulo 12. Indeed, if such a y existed, we could multiply both sides of the congruence by 2 to obtain

$$12y \equiv 10(\mod 12).$$

But $12y \equiv 0(\mod 12)$ so no such y can exist. Thus the correct answer is *none of the above*.

(c) (4 points) A value of z for which $z \equiv 54(\mod 3)$ and $z \equiv 55(\mod 4)$ is:

○ 0 ○ 1 ○ 2 ● 3 ○ 4 ○ 5 ○ 6 ○ 7
○ 8 ○ 9 ○ 10 ○ 11 ○ none of the above.

Solution: Since $54 \equiv 0(\mod 3)$ and $55 \equiv 3(\mod 4)$, the congruences are equivalent to

$$z \equiv 0(\mod 3) \qquad z \equiv 3(\mod 4).$$

We could solve this using the Chinese remainder theorem, or simply observe that $z = 3$ satisfies both congruences, so $z = 3$ is the correct answer (and the answer is unique modulo 12 by the Chinese remainder theorem).

6. (12 points) Prove or disprove: if x is a rational number and x^2 is an integer, then x must be an integer.

Solution: Proof: Assume x is rational and x^2 is an integer. Assume without loss of generality that x is positive (since the case $x = 0$ is trivial and otherwise we may consider $-x$ instead of x). We will show that x is an integer.

Choose positive integers p and q such that $x = p/q$ and p and q *have no common factors other than* 1. Assume for contradiction that $q > 1$. Since x^2 is an integer we have $q^2|p^2$ so $gcd(p^2, q^2) = q^2 > 1$. Note also that $p > 1$ since $q^2|p^2$.

We now observe that $gcd(p^2, q^2) > 1$ implies $gcd(p, q) > 1$: if instead $gcd(p, q) = 1$, the prime factorizations $p = p_1^{i_1} \dots p_m^{i_m}$ and $q = q_1^{j_1} \dots q_n^{j_n}$ would consist of disjoint sets of primes $\{q_1, \dots, q_n\}, \{p_1, \dots, p_m\}$, which would imply by the uniqueness part of the fundamental theorem of algebra that that the prime factorizations $p^2 = p_1^{2i_1} \dots p_m^{2i_m}$ and $q^2 = q_1^{2j_1} \dots q_n^{2j_n}$ also consist of disjont sets of primes, which would imply $gcd(p^2, q^2) = 1$.

The fact that $gcd(p, q) > 1$ contradicts our choice of p and q. Thus we must have $q = 1$, so $x = p/q$ is an integer, as desired.

This problem can be viewed as a generalization of the proof that $\sqrt{2}$ is irrational. For an alternate proof that more directly uses the fundamental theorem of algebra, see Prof. Haiman's Spring 2021 final at
https://math.berkeley.edu/ mhaiman/math55-spring21/.

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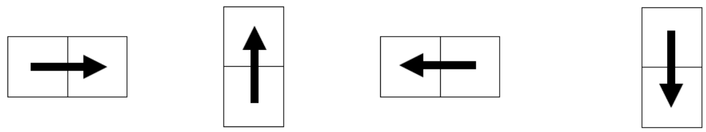
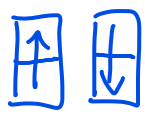
7. A *dominarrow* is a 2×1 tile labeled with an arrow, which may be oriented in four ways as follows:



A *tiling* of a $2 \times n$ grid is a placement of dominarrows on the grid such that each square is occupied exactly once. Two tilings are the same iff they contain exactly the same dominarrows in the same locations with the same orientations.

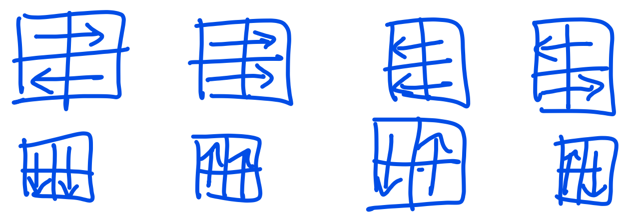
(a) (3 points) How many dominarrow tilings do the 2×1 grid and 2×2 grid have?

2×1:



$\therefore a_1 = 2$

2×2:



$\therefore a_2 = 8$

(b) (8 points) Let a_n be the number of tilings of the $2 \times n$ grid. Find a recurrence relation for a_n and prove your answer. Be sure to write down the base case(s).

Solution: Above, we have calculated the base cases $a_1 = 2$ and $a_2 = 8$.
For $n \geq 3$, in any tiling of a $2 \times n$ grid, the upper right grid square is occupied either by:

1. A vertical tile, oriented either up or down (2 such ways). In this case the remainder of the tiling is an arbitrary tiling of a $2 \times (n-1)$ grid. The total number of such tilings is therefore $2a_{n-1}$.

2. A horizontal tile, oriented either right or left (2 such ways). There must be a horizontal tile below this one, which may also be oriented right or left (2 ways for each choice of the first horizontal tile). The remainder of the tiling is an arbitrary tiling of a $2 \times (n-2)$ grid. The total number of such tilings is therefore $2 \times 2 \times a_{n-2}$.

Since the above possibilities are exhaustive and mutually exclusive, by the sum rule the total number of dominarrow tilings is

$$a_n = 2a_{n-1} + 4a_{n-2}.$$

This problem is a twist on a homework problem, 8.1#26, which considers tilings by unlabeled tiles (without arrows). An alternate way to do the problem is to *let b_n be the number of tilings of a $2 \times n$ grid by tiles without arrows, show that $b_n = b_{n-1} + b_{n-2}$, and observe that $a_n = 2^n b_n$ since every tiling of the $2 \times n$ grid has n tiles, each of which can have 2 possible directions of arrows.*

(c) (3 points) Using your answer to part (b), find a_4.

Solution: $a_3 = 2a_2 + 4a_1 = 16 + 8 = 24 \qquad a_4 = 2a_3 + 4a_2 = 48 + 4(8) = 80.$

8. (a) (2 points) Give an example of a graph on 4 vertices with degree sequence $3, 2, 2, 1$.

Solution: $G = (V, E)$ with $V = \{1, 2, 3, 4\}, E = \{12, 23, 13, 34\}$.

(b) (6 points) Suppose $G = (V, E)$ is a connected simple graph on n vertices with exactly n edges. Show that there is an edge $e \in E$ such that $G' = (V, E - \{e\})$ is a tree.

Solution: Since $G = (V, E)$ is connected, we can choose a spanning tree $T = (V, F)$ of T. A tree on n vertices has exactly $n - 1$ edges. Choose e to be the unique edge in $E - F$ (which must exist since $|E| = n$ and $|F| = n - 1$). Now the graph $G' = (V, E - \{e\})$ is a tree, as desired.

(c) (6 points) Suppose G is a connected simple graph on n vertices with exactly n edges. Show that G is $3-$colorable.

Solution: Suppose $G = (V, E)$ is a connected simple graph on n vertices with n edges. By part (a), we can choose an edge $e = xy$ such that $G' = (V, E - \{e\})$ is a tree. We showed in class that every tree is $2-$colorable, so we can choose a $2-$coloring $f : V \to \{1, 2\}$ of G'. Notice that $f(u) \neq f(v)$ for every edge $uv \in E - \{e\}$, so f is almost a $2-$coloring of G except that it might assign the same color to x and y. Define a new coloring $g : V \to \{1, 2, 3\}$ by $g(u) = f(u)$ for all $u \neq x$ and $g(x) = 3$. Notice that $g(u) \neq g(v)$ for all $uv \in E - \{e\}$ and $g(x) \neq g(y)$ since $g(y) \in \{1, 2\}$ and $g(x) = 3$. Thus g is a $3-$coloring of G, as desired.

Graph Theory Definitions

A **graph** is a pair $G = (V, E)$ where V is a finite set of **vertices** and E is a finite multiset of $2-$element subsets of V, called **edges**. If E has repeated elements G is called a **multigraph**, otherwise it is called a **simple graph**. Two vertices $x, y \in V$ are **adjacent** if $\{x, y\} \in E$. An edge $e = \{x, y\}$ is said to be **incident with** x and y. The **degree** of a vertex x is the number of edges incident with it.

A **path** of length k in G is an alternating sequence of vertices and edges

$$x_0, e_1, x_1, e_2, \ldots, x_{k-1}, e_k, x_k$$

where $e_i = \{x_{i-1}, x_i\} \in E$ for all $i = 1 \ldots k$. The path is said to **connect** x_0, x_k, **visit** the vertices $x_0, \ldots, x_k$, and **traverse** the edges $e_1, \ldots, e_k$. A path is called **simple** if it contains no repeated vertices. If $x_0 = x_k$ then the path is called a **circuit**. A graph is **connected** if it contains a path between every pair of distinct vertices.

A **k-coloring** of G is a function $f : V \to \{1, \ldots, k\}$ such that $f(x) \neq f(y)$ whenever x is adjacent to y. A graph is $k-$**colorable** if it has a k-coloring.

A graph $H = (W, F)$ is a **subgraph** of G if $W \subseteq V$ and $F \subseteq E$. If $W = V$ then H is called a **spanning** subgraph of G. A **connected component** of G is a maximal connected subgraph of G, i.e. a subgraph $H \subseteq G$ such that H is connected and adding any vertices or edges to H would make it disconnected.

A **cycle** is a circuit with no repeated edges and no repeated vertices, except the end-points. A graph is called **acyclic** if it does not contain a cycle as a subgraph. A connected acyclic graph is called a **tree**.

An **Eulerian circuit** of a connected multigraph G is a circuit which traverses every edge of G exactly once.